

THE DOUBLE LADDER THEOREM — LEAN CERTIFICATE

A machine-checked proof, in Lean 4 with Mathlib, of the structure of the discriminant group of the lattice of lines on the Fermat surface of prime degree

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Certificate v2 · 30 September 2026. v2 corrects v1 (the same day) after a cold read by an independent reader, who found no fatal error and no gap: four wrong statements and several unclear ones in the text of the certificate, listed in §11. The Lean content, the numbers and the fingerprints are unchanged.

Companion to *The Double Ladder Theorem* (R. Amichis Luengo, 5 June 2026, version 4 of 30 September 2026), file `hodge-fermat-campaign/THE_DOUBLE_LADDER_THEOREM.md` of `github.com/tretoef-estrella/chaise-longue-theorem` and `papers/THE_DOUBLE_LADDER_THEOREM.md` of `github.com/tretoef-estrella/watermark-theorem`; to the proof that the Lean formalization follows, [DL-Pencil]; and to the *Lean Certificate of The Watermark Theorem* [WM-Lean], on which it builds. The Lean project and all the evidence are in the repository *watermark-theorem*, folder `lean/`.

0. Verdict

Let m be a prime and V the lattice spanned by the $3m^2$ lines of the Fermat surface of degree m , with the bilinear form given by their intersection numbers. Its discriminant group is $D = V^*/V$, where $V^* = \text{Hom}(V, \mathbb{Z})$ and V sits in V^* through the form. Lean 4 proves:

- for every prime $m \geq 7$: $D \cong (\mathbb{Z}/m)^{3m^2-24m+59} \times (\mathbb{Z}/m^2)^{3m-16}$;
- for $m = 5$: $D \cong (\mathbb{Z}/5)^{10} \times \mathbb{Z}/25$.

These are exactly the statements of the paper. The order of D , $m^{3(m-3)^2}$, is the Watermark Theorem, certified in Lean in [WM-Lean]. This certificate adds its *structure*: the exponent divides m^2 , and the number of cyclic factors of order m^2 is $3m - 16$ (1 at $m = 5$).

The whole proof compiles on the author's Mac, including the case $m = 5$. It contains no `sorry`, no `admit`, no `native_decide` and no added axiom. Both final theorems depend only on the three standard axioms of Lean and Mathlib: `propext`, `Classical.choice` and `Quot.sound`.

What Lean certifies, and what it does not. As for the Watermark, Lean works with the explicit integer matrix G of size $3m^2 \times 3m^2$ defined in `Watermark/Defs.lean`, and with $V = \mathbb{Z}^{3m^2}/\ker G$ carrying the form induced by G . That G is the intersection matrix of the lines on the complex Fermat surface is geometry: it is proved by hand and checked by computer in [WM-Lean, §1.4], and it is not formalized. That V is the whole Néron-Severi group for prime $m \geq 5$ is a cited theorem [SSvL, Deg]. So:

Lean certifies the lattice statement completely. The statement about the Néron-Severi group of the Fermat surface follows from it by the identification of [WM-Lean, §1.4],

proved by hand and checked by computer, and by the cited results of [SSvL] and [Deg].

The paper's own proof is not the one formalized. The June paper proves the theorem by characters, shadows and a «rank ladder». Four steps there, in its §4 and §6, were confirmed at three or four primes and not written for general m (its version note lists them). The Lean proof follows a different, shorter route, written for general m [DL-Pencil] (§3 below), which does not need those steps.

§1 states exactly what is proved. §2 states exactly what is not. §3–§8 describe how it was proved and how to check it. §9 records what was already known.

1. The certified statement

1.1 The theorems, verbatim

Printed by Lean with `#check` and `#print` `axioms` (`log check_run3_clean.log`, after a clean rebuild of the whole project):

```
double_ladder_theorem : ∀ (m : ℕ) [inst : Fact (Nat.Prime m)],
  7 ≤ m → Nonempty (D m ≃+ (Fin (3 * m ^ 2 + 59 - 24 * m) → ZMod m) × (Fin (3 * m - 16) → ZMod (m ^ 2)))

double_ladder_five : Nonempty (D 5 ≃+ (Fin 10 → ZMod 5) × ZMod 25)
```

```
'DoubleLadder.double_ladder_theorem' depends on axioms: [propext, Classical.choice, Quot.sound]
'DoubleLadder.double_ladder_five' depends on axioms: [propext, Classical.choice, Quot.sound]
```

$\simeq+$ is an isomorphism of additive groups. $\text{Fin } n \rightarrow \text{ZMod } k$ is the group $(\mathbb{Z}/k)^n$.

1.2 The definitions it uses, verbatim

The statement uses one new definition, in `RequestProject/DoubleLadder/DL3.lean`, printed by `#print` in `check_run3_clean.log`:

```
@[reducible] def DoubleLadder.D : (m : ℕ) → [NeZero m] → Type :=
fun m [NeZero m] => Module.Dual ℤ (V m) / LinearMap.range (formV m)
```

Everything else — *Line*, *gram*, K , V , `formV` — is the Watermark's, in `RequestProject/Watermark/Defs.lean`. It is printed verbatim in [WM-Lean, §1.2], and its faithfulness to the lines of the Fermat surface is audited there line by line [WM-Lean, §1.3]. The Watermark files used here are **byte-identical** to those certified in [WM-Lean]: their SHA-256 fingerprints, listed in §8, are the same as in [WM-Lean, §8].

`formV m` is a bilinear form on $V m$, that is, a linear map $V m \rightarrow \text{Module.Dual } \mathbb{Z} (V m)$. Its range is the image of V in V^* . So $D m$ is V^*/V , the discriminant group of the lattice V .

1.3 Faithfulness to the paper, line by line

The paper is *The Double Ladder Theorem*, Abstract and §1. The statement is the same in versions 2 (5 June 2026, md5 7b10b75d...), 3 and 4 (30 September 2026); version 4 is the one published with this certificate.

Paper	Lean	Agreement
S_m , prime $m \geq 7$	[Fact (Nat.Prime m)], $7 \leq m$	Identical, and no other hypothesis.

Paper	Lean	Agreement
$m = 5$	double_ladder_five, stated at $m = 5$	Identical.
$V = NS(S_m)$, the lattice of the lines	$V\ m$ of the Watermark, with fo rmV m	The lattice of the lines, as in [WM-Lean, §1.3–§1.4]. $V = NS$ is cited, §2.
the discriminant group V^*/V	$D\ m = \text{Module.Dual } \mathbb{Z}$ $(V\ m) / \text{range } (\text{formV}$ $m)$	Identical (§1.2).
$\cong$	Nonempty ($D\ m \simeq + \dots$)	An isomorphism of abelian groups exists.
$(\mathbb{Z}/m)^{3m^2-24m+59}$	$\text{Fin } (3 * m ^ 2 + 59 -$ $24 * m) \rightarrow \text{ZMod } m$	Identical. Subtraction in $\mathbb{N}$ does not truncate: $3m^2 + 59 - 24m$ is written with the subtraction last, and $3m^2 - 24m + 59 > 0$ for every m (its discriminant is $576 - 708 < 0$).
$(\mathbb{Z}/m^2)^{3m-16}$	$\text{Fin } (3 * m - 16) \rightarrow \text{ZM}$ $\text{od } (m ^ 2)$	Identical. No truncation: $3m - 16 \geq 5$ for $m \geq 7$.
$(\mathbb{Z}/5)^{10} \times \mathbb{Z}/25$	$(\text{Fin } 10 \rightarrow \text{ZMod } 5) \times \text{Z}$ $\text{Mod } 25$	Identical.

Consistency with the Watermark. The order of the group in the statement is $m^{3m^2-24m+59} \cdot m^{2(3m-16)} = m^{3m^2-18m+27} = m^{3(m-3)^2}$, and $5^{10} \cdot 25 = 5^{12}$: the order certified in [WM-Lean]. This is also how Lean proves it (§3.1, step 4).

Non-vacuity.

- The hypotheses can be satisfied, and the statement is a concrete isomorphism, not a statement about an empty object. The file `checks_lean/CheckDLFinal.lean` instantiates the theorem at $m = 7, 11, 13$ and obtains, in Lean:
 - $D\ 7 \simeq + (\mathbb{Z}/7)^{38} \times (\mathbb{Z}/49)^5$;
 - $D\ 11 \simeq + (\mathbb{Z}/11)^{158} \times (\mathbb{Z}/121)^{17}$;
 - $D\ 13 \simeq + (\mathbb{Z}/13)^{254} \times (\mathbb{Z}/169)^{23}$.

The file compiles with no error: exit 0, 99 s, peak 2.20 GB (`checkdlfinal.log`, 30 Sep, after the clean rebuild). The same run prints the two final theorems with their axioms, D , Y , and four intermediate theorems (`gram_mul_ylad_mul_gram`, `exists_formV_eq_smul`, `rank_gram_map`, `rank_gram_map_five`), with the axioms of three of them (all but `exists_formV_eq_smul`). The axioms of every theorem in the cone are contained in those of the final theorems, which are `[propext, Classical.choice, Quot.sound]`.

- These three profiles, and the profile at $m = 5$, are the rows (2,5), (2,7), (2,11), (2,13) of Table 1 of Aljovin–Movasati–Villaflor [AMV], who computed them by machine, one cell at a time (§9).

2. What is not certified

1. The geometry.

- That G is the intersection matrix of the lines on S_m : proved by hand and checked by computer at $m = 5, 7, 11$ in [WM-Lean, §1.4]; not formalized.
- That $A/\ker G$ is the lattice of lines in $H^2(S_m, \mathbb{Z})$: standard, cited in [WM-Lean, §1.4].

Mathlib has no complex algebraic surfaces and no singular cohomology of them.

1. **The equality $V = NS(S_m)$** for prime $m \geq 5$: [SSvL] for $m \leq 100$, [Deg] for every m prime to 6; cited. With it, the theorem gives the discriminant group of $NS(S_m)$.
2. **The corollaries of the paper's §8** are not formalized.
 - The transcendental lattice T_m has the same discriminant group, because $H^2(S_m, \mathbb{Z})$ is unimodular. This is standard lattice theory [Nik], applied to a geometric input; it is not formalized.
 - The Brauer group consequences are not formalized.
3. **The paper's own proof is not the proof formalized.** Lean follows the route of §3.1, which is that of [DL-Pencil]: the same three facts and the same assembly. Three details differ, listed in [DL-Pencil, §7]; the main one is that Lean does not take a Smith form but counts $|D|$ and $|D/mD|$ (step 4 of §3.1). It is a second, independent proof of the same statement.
4. **Faithfulness is an audit, not a theorem.** Lean guarantees that the proof is correct for the statement as written. That the statement says what the paper says is checked in §1.3. A reader should check §1.2–§1.3 here and in [WM-Lean]: the definition D above, the file `Defs.lean`, and six cases of `gramUpper`.
5. **Timing data.** Aristotle's own start and end times were not exported. The times in §6 are the download times of the result files.

3. How the proof is organised in Lean

The proof was cut into three pieces, DL1–DL3, sent to Aristotle in the Watermark's project (the Double Ladder uses the Watermark's definitions and theorems). Each piece is a self-contained Markdown file in `pieces/` with Setting, Statements, Proof and Checks, and produced one Lean file in `RequestProject/DoubleLadder/`. No piece modified a file of an earlier piece: the Watermark files `W1–W7`, `Defs`, `Main`, `lakefile.toml`, `lean-toolchain` and `lake-manifest.json` were byte-identical in every returned project.

#	Piece	Content	Lean file	Main Lean results
DL1	<code>dl1_exp onent.m d</code>	the certificate matrix Y ; $GYG = 6m^2G$; $m^2V^* \subseteq V$	<code>DL1.lean</code>	<code>ylad, gram_mul_ylad_mul_gram, exists_toMa trix_mul_eq, exists_formV_eq_smul</code>
DL2	<code>dl2_ran k_mod_ m.md</code>	the shadow factorization $G = S^T \Gamma S - mI$; the 6 relations per family; $\text{rank}(G \bmod m) = 12(m-3)$ for $m \geq 7$; 26 at $m = 5$	<code>DL2.lean</code>	<code>gram_eq_shadow, ker_relMap_eq_span, fin rank_ker_relMap, sum_mul_shift_eq_zero, rank_gram_map, rank_gram_map_five</code>
DL3	<code>dl3_ass embly.m d</code>	$D \cong \mathbb{Z}^r / \text{range } M$; $ D = \det $; $m^2D = 0$; $ D/mD = m^{r - \text{rank } \tilde{M}}$; the group-structure lemma; the two final theorems	<code>DL3.lean</code>	<code>discrEquiv, card_coker, sq_smul_discr_e q_zero, card_coker_quot_smul, equiv_of_ card, double_ladder_theorem, double_lad der_five</code>

3.1 The chain of the final theorem

This is how `double_ladder_theorem` is proved in Lean. Every step is a Lean proof. Here $m \geq 7$ is prime, G is the Watermark's matrix, $r = \text{rank } V = 3(m-1)(m-2) + 1$.

1. **The order** (Watermark, W7). For any $\mathbb{Z}$ -basis of V , $\det \text{Gram} = m^{3(m-3)^2}$. In DL3 this becomes $|D| = m^{3(m-3)^2} (\text{card_coker, via Mathlib's Submodule.natAbs_det_equiv})$.

2. **The exponent** (DL1). Let Y be block diagonal over the three families, with $Y = P_k + P_l + 3P_{k-l} + 3P_{k+l} - 6m \cdot I$ on each family, where $P_\pi(p, q) = 1$ when the pencil π takes the same value at p and q . Then $\mathbf{G} \mathbf{Y} \mathbf{G} = 6m^2 \mathbf{G}$ over Z . It is proved on the pullback vectors of W4 and the family vectors. Y acts on each of them by a scalar. So does G , except on the overlap pullbacks, which come in pairs v, v' with $Gv = -m v + m v'$; Lean treats each pair jointly (defect_pair). It follows that $6m^2 \cdot V^* \subseteq V$. Since $|D|$ is a power of the prime $m \geq 5$, 6 is invertible on D , so $m^2 \cdot D = 0$ (exists_formV_eq_smul, sq_smul_discr_eq_zero).

3. **The rank modulo m** (DL2).

- Modulo m , G factors through twelve «shadows»: for each family and each pencil $\pi \in \{k, l, k-l, k+l\}$, the map that sums a function over the fibres of π . Over Z , $\mathbf{G} = \mathbf{S}^T \mathbf{T} \mathbf{S} - \mathbf{m} \cdot \mathbf{I}$, with Γ an explicit symmetric matrix on the twelve shadow slots (gram_eq_shadow).
- The annihilator of the image of S is spanned by six relations per family (three constants, two linear, and the polarization identity $(k+l)^2 + (k-l)^2 = 2k^2 + 2l^2$), and no others (ker_relMap_eq_span, finrank_ker_relMap). Lean proves this with mixed second differences of functions on Z/m (LemmaC.secondDiff), not by writing functions as polynomials of degree $\leq m-1$ as the pencil proof does; the statement is the same.
- The image of Γ on the relations lies back in the image of S when $m \geq 7$, because every pairing involved is $\sum_{x \in F_m} p(x)q(x)$ with $\deg(pq) \leq 4 \leq m-3$, and $\sum_x x^s = 0$ for $s \leq m-2$ (sum_mul_shift_eq_zero).
- Together: $\text{rank}_{F_m}(\mathbf{G} \bmod m) = 12(m-3)$ (rank_gram_map).

4. **The assembly** (DL3). D/mD has $m^r - \text{rank}(\mathbf{G} \bmod m)$ elements (card_coker_quot_smul). A finite abelian group H with $m^2 H = 0$, $|H| = m^N$ and $|H/mH| = m^c$ is $(Z/m)^{2c-N} \times (Z/m^2)^{N-c}$ (equiv_of_card, a general lemma). Here $N = 3(m-3)^2$ and $c = r - 12(m-3) = 3m^2 - 21m + 43$, so $N - c = 3m - 16$ and $2c - N = 3m^2 - 24m + 59$.

The case $m = 5$ (double_ladder_five). Steps 1, 2 and 4 are the same. Step 3 changes, because $\sum_{x \in F_5} x^4 = -1 \neq 0$: the degree-4 pairings survive and the rank is 26, not $12(5-3) = 24$. Lean proves $\text{rank}(\mathbf{G} \bmod 5) = 26$ by two explicit certificates over $Z \bmod 5$, checked by evaluation in the kernel (decide + kernel): a factorization $\bar{G} = B \cdot C$ through 26 columns (so the rank is at most 26), and a 75×26 matrix Y_5 such that 26 chosen rows of $\bar{G}$ times Y_5 give the identity I_{26} (so the rank is at least 26). Then $c = 37 - 26 = 11$, $N = 12$, $N - c = 1$, $2c - N = 10$.

3.2 The dependency cone

The Lean program checks_lean/DepsDL.lean collects every declaration of the project that double_ladder_theorem and double_ladder_five use, directly or indirectly.

Result (deps_dl.log, exit 0, 106 s):

Module	double_ladder_theorem	double_ladder_five
Watermark.Defs	38	38
Watermark.W1	34	34
Watermark.W2	81	81
Watermark.W3	38	38
Watermark.W4	31	31
Watermark.W5	50	50
Watermark.W6	103	103

Module	double_ladder_theorem	double_ladder_five
Watermark.W7	44	44
DoubleLadder.DL1	47	47
DoubleLadder.DL2	96	125
DoubleLadder.DL3	47	41
Total	609	632

Every file of the Watermark and of the Double Ladder is on the path of both final theorems; only the header `Main.lean` (no declarations) is not. Every one of these declarations is checked by the kernel as part of the final theorems. For $m = 5$ the cone has 29 more declarations of DL2 (its explicit certificates) and 6 fewer of DL3 than for $m \geq 7$. These are differences of counts; the two cones were not compared declaration by declaration.

4. Trust base

- **What must be trusted:**
 - the Lean 4 kernel, toolchain `leanprover/lean4:v4.28.0`;
 - Mathlib at commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365` (tag `v4.28.0`);
 - the three standard axioms `propext`, `Classical.choice`, `Quot.sound`;
 - the audit of the statement (§1.3 here and in [WM-Lean]) and the geometry of [WM-Lean, §1.4].
- **What need not be trusted:** Aristotle (Harmonic), the AI system that wrote the Lean proofs, and the author's brute-force checks. Every file was compiled again on the author's machine, and the kernel re-checks every proof.
- **Forbidden constructs, searched as whole words in the three Double Ladder files:** none of `sorry`, `admit`, `axiom`, `native_decide`, `implemented_by`, `extern`, `unsafe`, `opaque`, `csimp`, `partial`, `macro`, `elab`, `syntax`, `notation`, `debug` occurs. (For the nine Watermark files, see [WM-Lean, §4]; they are unchanged.)
- **decide.** It occurs, and every occurrence is kernel evaluation, not native code:
 - `DL1.lean`, once: `interval_cases m <=> ... <=> decide`, on small numbers;
 - `DL2.lean`: the $m = 5$ certificates, *decide + kernel*, 41 small lemmas, plus *by decide* for indices $\langle 0, \dots, \text{by decide} \rangle : \text{Fin } 3$.

decide + kernel asks the kernel itself to evaluate a decidable proposition. It adds nothing to the trust base, unlike `native_decide`, which does not occur.

- **Options.** `DL2.lean` sets `maxRecDepth 100000` on 43 declarations (the $m = 5$ certificates and their assembly). This raises an elaboration limit; it does not affect soundness. `DL1.lean` and `DL3.lean` set no option.
 - **Axioms of every main theorem**, printed after each piece (`check_run1.log`, `check_run2_general.log`, `check_run3.log`, `check_run3_clean.log`): `always [propext, Classical.choice, Quot.sound]`.
-

5. The procedure

The procedure is that of [WM-Lean, §5] and of the Chaise Longue certificate [CL-Lean]: write a self-contained piece; check it by brute force before sending, with a negative control; send it to Aristotle (project 884019b5-ec06-4ff7-8a7b-8d8a1f18d0a1); reload the project page and confirm the task finished before downloading; diff against the local project; grep for forbidden constructs; build locally inside a watchdog; print statements and axioms; audit faithfulness; record the run in ESTADO_DOUBLE_LADDER.md.

Stop conditions, fixed in advance: a property NEGATED by Aristotle; a statement that is not faithful; a build failure; a sorry or an extra axiom; a change to an old file. **None occurred.**

The one local change to Aristotle's output, stated in full.

- Aristotle's DL2.lean proved each of the two $m = 5$ certificates as **one** *decide +kernel* theorem: hBC (75 rows) and hXY (26 rows). On Aristotle's machine this took about 2 minutes.
- On the author's Mac (8 GB) the local build did not finish. The kernel's cache grows within a single declaration: one row costs about 2 s and no memory; 25 rows in one theorem cost about 49 s and 1 GB more (logs test5_*.log). We estimate, from these measurements, that the whole certificate in one theorem needed about 9 GB, so the machine swapped; no log records that figure. The first watchdog measured only resident memory and did not see it; the watchdog now measures the real footprint.
- **Fix:** hBC was split into 15 lemmas, one per *(family, a)*, and hXY into 26 lemmas, one per row. Each is still *decide +kernel*, and hBC and hXY are then assembled from them by *ext* and *fin_cases*.
- **The statements of hBC, hXY and rank_gram_map_five, the whole general part of DL2 and both final theorems are byte-identical to Aristotle's.** Only the proofs of hBC and hXY changed. The complete diff against Aristotle's file is DL2_split_m5.diff (four hunks, all inside namespace DL2.Five); Aristotle's original is kept in run2/.
- With the split, the full DL2.lean builds in 86 s.

Other incidents.

- DL1: the build and the checks passed on the first run.
- DL2: before the split, the general part (everything except namespace DL2.Five) was certified on its own (checks_lean/DL2_general_only.lean, Aristotle's text verbatim up to that namespace), 149 s, axioms standard.
- DL3: no kernel evaluation of its own; built in 29 s.

6. Data per piece

#	Aristotle task	Tarball	md5	Downloaded	Build (s)	Peak (GB)	Lines	Thms	Defs	Checks before sending
DL1	4bc5652a	(8)	42ae52ff	30 Sep 09:18	72	1.59 ¹	394	35	4	$G Y G = 6m^2 G$ exact at $m = 5, 7, 11, 13, 17$; control $G Y G = 6m \cdot G$ fails at all five; minimal polynomial $x(x+m)(x+2m)(x-3m)$ at all five

#	Aristotle task	Tarball	md5	Downloaded	Build (s)	Peak (GB)	Lines	Thms	Defs	Checks before sending
DL2	f130f94c	(9)	429bc685	30 Sep 09:55	86 ²	4.15	1079 ³	71	24	$G = S^\Gamma \Gamma S - mI$ at $m = 5, 7, 11, 13$; $\dim \text{im } S = 12m - 18$; $\dim(W \cap W^\perp) = 18$ at $7, 11, 13$ and 16 at 5 ; $\text{rank}(G \bmod m) = 26, 48, 96, 120$; control: three degree-4 pairings nonzero at $m = 5$
DL3	24f74a8c	(10)	11b808fe	30 Sep 10:42	29	3.54	415	15	2	Smith profiles of G over $\mathbb{Z}_m$: $\{26, 10, 1\}, \{48, 38, 5\}, \{96, 158, 17\}, \{120, 254, 23\}$ at $m = 5, 7, 11, 13$; $9m - 7$ divisors of valuation ≥ 4 (the kernel) in each case

¹ Measured by the first watchdog, resident memory only; the later figures are the real footprint. ² With the $m = 5$ certificates split (§5). Aristotle's unsplit file was stopped by the watchdog at 901 s. ³ Local file, with the split; Aristotle's file has 774 lines.

Tarballs: files 884019b5-ec06-4ff7-8a7b-8d8a1f18d0a1-aristotle (N).tar.gz, extracted to run1/, run2/, run3/. Lines, theorems and definitions are counted as in [WM-Lean, §6].

7. Totals

Quantity	Value
Pieces sent to Aristotle	3 (one task each)
Lean files of the Double Ladder	3 (DL1, DL2, DL3), on top of the 9 files of the Watermark
Lines of Lean	1 888 (local; with Aristotle's unsplit DL2, 1 583)
Theorems and lemmas	121
Definitions	30
sorry / admit / added axioms / native_decide	0 / 0 / 0 / 0
Axioms of the two final theorems	propext, Classical.choice, Quot.sound
Calendar time	30 Sep 2026: first result downloaded 09:18 (DL1), last 10:42 (DL3); clean rebuild and final check the same day
Clean rebuild of the whole project (Watermark + Double Ladder, 11 modules)	exit 0, 333 s , peak 4.14 GB (build_clean_all.log). The log keeps only the last lines of the build output and the watchdog's closing line: the 11 project modules Defs, W1–W7, DL1–DL3 each rebuilt, then «Build completed successfully (8036 jobs)». The command line itself is not in the log. The header Main.lean is not rebuilt because nothing imports it.
Final check after the clean rebuild	121 s, peak 2.52 GB, axioms standard (check_run3_clean.log)

8. Reproducibility

Machine: Apple Mac (arm64), 8 CPU cores, 8 GB of memory, macOS (Darwin 25.5.0). **Toolchain:** lean-prover/lean4:v4.28.0; Mathlib v4.28.0 = commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365. The same toolchain and Mathlib as [WM-Lean] and [CL-Lean].

Commands, from the project root:

```
lake exe cache get
lake build # all files; exit 0
lake env lean <path>/checks_lean/CheckDLFinal.lean # statements, axioms, D, Y, non-vacuity
lake env lean <path>/checks_lean/DepsDL.lean # the dependency cone
```

Memory. Loading Mathlib alone takes about 3 GB. The full build peaks at about 4.2 GB, in DL2.lean. A machine with 8 GB builds it; with less, expect swapping.

Fingerprints of the source (the 12 .lean files of the project). They were computed on 30 September, after the final check, on the files that were built; the build and check logs themselves do not record them.

- SHA-256 of the concatenation, in sorted path order: 02ca3b3d8b06e9cc1a2db-d4827365bed630f30cbe1747eb7f8e064bde427f1aa.
- SHA-256 of the list of individual SHA-256 lines (shasum -a 256 \$(find RequestProject -name '*.lean' | sort) | shasum -a 256): bb612c10e71963482901dff9ce489755a6cf855df4cce-ba241ec564951fcd8b0.

File	SHA-256
RequestProject/DoubleLadder/DL1.lean	4572065d4a5f25a147c94bca0dc9ad01f33bd848775946280ce6e794079bb707
RequestProject/DoubleLadder/DL2.lean	7e7173b87edbd78e4f2b5273caa17c50988145f7b21eca9d05606af8f5ddc695
RequestProject/DoubleLadder/DL3.lean	17bf23320835c2ee9c4b07e8bd36024b9c7ad8fe0c8986f814572633ac1925a4
RequestProject/Main.lean	929b0bddef0b781f3fb42c7a99f252dc0bda7331f698104f7075e12ff637c52d
RequestProject/Watermark/Defs.lean	0af5ba018edd1a64138a40349e3d448062cdb20dbb5c2634198f47ff19af12b5
RequestProject/Watermark/W1.lean	fe5a814f4feba342697dd66dcaa9d95d403a06d4e458afc9014e08664649075a
RequestProject/Watermark/W2.lean	29580b50a72a8b11bf02c2f622198e632ff92b0df4386ebe4fd5dc46395c9342
RequestProject/Watermark/W3.lean	61a2e26993834c755c580c39dd01b0b936f5657eeef8f88c4c4208a9ffbefbd7
RequestProject/Watermark/W4.lean	a2419a27d3bf7d6c14e94135258f51271142b7049695ed379c84e78553077a06
RequestProject/Watermark/W5.lean	05a5311a1761f00fc1c260548d0d6f43a981384abe286a59f1ec3a2d42eb4858
RequestProject/Watermark/W6.lean	f0fbd352eee1f075ec7640643fe68b0eb8303ef4404be7a4155644309b734839

File	SHA-256
RequestProject/Watermark/W7.lean	bc93a6455c604248ebfa186150b7f6eb66662c0c65c4ad1738df8d499f473a58

The nine Watermark rows are identical to [WM-Lean, §8].

On the author's machine (folder ARISTOTLE_LEAN/DOUBLE_LADDER/ of the author's archive; the Lean project itself is ARISTOTLE_LEAN/WATERMARK/proyecto/): the three pieces; run1/–run3/; the brute-force scripts and logs in checks/; the Lean check files in checks_lean/; every build and check log; DL2_split_m5.diff; the state file ESTADO_DOUBLE_LADDER.md; and the pencil proof [DL-Pencil]. **Where the files are published:** the repository github.com/tretoef-estrella/watermark-theorem, folder lean/; the project is *lean/project/*, and everything listed here is in *lean/double-ladder/*, with the same names and relative paths (the returned projects run1/–run3/ are not published, and neither is one empty log of the interrupted build).

Recommended to a reader who wants to remove the remaining trust in the author's logs: an independent lake build on another machine, and a run of lean4checker. Neither has been done yet.

9. What was known before

This section records what we found in the literature on 30 September 2026. A search cannot prove that nothing exists; it records what we looked at. Sources read in the original are marked «read».

- **Shioda (1987)** [Sh87], read (Project Euclid, 17 pages). §7, p. 133. *Question 7.2:* for the Fermat surface of prime degree m in characteristic $p \equiv 1 \pmod{m}$, is $|\det NS(X)| = m^{3(m-3)^2}$? *Question 7.4 (i):* for the complex Fermat surface of prime degree m , are (7.10) $|\det NS| = m^{3(m-3)^2}$ and (7.11) « NS is spanned by the classes of lines» true? He asks for the determinant, not for the group. «Added in proof» (1): the discriminant divides a power of m for every m . The Watermark Theorem answers (7.10) for every prime $m \geq 5$, and [Deg] answers (7.11). The Double Ladder Theorem goes beyond the question: it gives the group, not only its order.
- **Schütt, Shioda, van Luijk (2010)** [SSvL], read (arXiv source, dated 30 Sep 2009). End of §4: for every odd $m \leq 81$, the determinant of the intersection form of the lines of their basis $\mathcal{B}$ is $m^{3(m-3)^2}$, «with exponent as conjectured in [Sh87]». They prove that the lines span $NS \otimes \mathbb{Q}$, and span NS for $m \leq 100$, $\gcd(m, 6) = 1$. They give the determinant, not the group.
- **Degtyarev (2015)** [Deg], read (arXiv source). For every $m \leq 4$ or $\gcd(m, 6) = 1$, the lines generate $NS(S_m)$ over $\mathbb{Z}$. The proof is topological; it gives no discriminant.
- **Shioda (2015)** [Sh15], read. In §6.5, Conjecture 23, for m prime to 6, he writes $\det NS(X_m) = m^{3(m-3)^2}$ inside a conjecture on Mordell–Weil lattices. So in 2015 the formula was still used as expected, not as proved.
- **Aljovin, Movasati, Villaflor (2019)** [AMV], read. Table 1 lists, by machine (Smith normal form, one cell at a time), the elementary divisors of the lattice of linear cycles of the Fermat surface for $3 \leq m \leq 14$. The rows for prime m are:
 - (2,5): $1^{26} \cdot 5^{10} \cdot 25$ — that is, $(\mathbb{Z}/5)^{10} \times \mathbb{Z}/25$;
 - (2,7): $1^{48} \cdot 7^{38} \cdot 49^5$;
 - (2,11): $1^{96} \cdot 11^{158} \cdot 121^{17}$;
 - (2,13): $1^{120} \cdot 13^{254} \cdot 169^{23}$.

All four agree with the Double Ladder Theorem. [AMV] give no general formula and no proof for general m . (Their sign column reads $-$ for $(2,7)$. For the lattice of the lines, Lean proves the sign $+$ (the Watermark), as the Hodge index theorem predicts: the rank minus one, 90, is even. The sign does not affect the group.)

- **Jumagulov (2026)** [Jum], read: Galois-invariant Néron–Severi ranks of Fermat surfaces; no discriminant.
- **Also screened:** the 51 works that Semantic Scholar lists as citing [SSvL] (titles and, where relevant, abstracts), and web searches for the formula, for the discriminant group of NS of Fermat surfaces, and for its elementary divisors. We found no general formula for the discriminant group, and no proof of the determinant formula, before June 2026.

Therefore. Before this work, the determinant $m^{3(m-3)^2}$ was conjectured (Shioda) and checked by computer for odd $m \leq 81$ [SSvL]; the discriminant group was known by computer for $m \leq 14$ [AMV]. As far as we could determine, a formula for all primes and its proof are new: the Watermark Theorem (the order) and the Double Ladder Theorem (the structure). Both are now certified in Lean for every prime $m \geq 5$.

Three corrections to earlier texts, found while writing this section.

1. *The Double Ladder Theorem* v2 (§1 and §9) says that the $m = 11$ profile lies «outside every published table» and that [AMV] supplies two of its three anchors. In fact [AMV, Table 1] lists all three, $m = 5, 7, 11$, and also $m = 13$. The $m = 11$ value was indeed predicted before it was computed, but it was not new. The theorem is unaffected.
2. The pencil proof, version 1 (§5 and §6), calls $m = 13$ «a new cell». It is row $(2,13)$ of [AMV]. Version 2 [DL-Pencil] corrects it.
1. *The Double Ladder Theorem* v2 (Abstract and §1) says that Shioda asked for the determinant «and, behind it, the discriminant form», and that the two theorems answer his Question 7.4 «in full ... not only the determinant ... but its discriminant form's underlying group». [Sh87] asks only for the determinant (Questions 7.2 and 7.4). The Watermark answers (7.10); the Double Ladder goes further.

10. What this certificate claims, in one paragraph

A computer proof assistant, Lean 4 with the Mathlib library, has checked a complete proof of the following statement, with no gap and no axiom beyond the three standard ones:

Let G be the explicit $3m^2 \times 3m^2$ integer matrix of [WM-Lean, §1.2], $V = Z^{3m^2}/\ker G$ with the form induced by G , and $V^* = \text{Hom}(V, Z)$. For every prime $m \geq 7$, $V^*/V \cong (Z/m)^{3m^2-24m+59} \times (Z/m^2)^{3m-16}$; and $V^*/V \cong (Z/5)^{10} \times Z/25$ for $m = 5$.

G is the intersection matrix of the $3m^2$ lines on the Fermat surface S_m ([WM-Lean, §1.4]: proved by hand, checked by computer, not formalized), and by [SSvL] and [Deg] $V = NS(S_m)$ for prime m . The statement is therefore the structure of the discriminant group of the Néron–Severi lattice of the Fermat surface of every prime degree $m \geq 5$. Together with the Watermark Theorem, certified in [WM-Lean], it gives the complete elementary-divisor profile of that lattice.

The proofs in Lean were written by the AI system Aristotle (Harmonic) from pieces written by the author. The author's audit, assisted by Claude (Anthropic), checked that the statements match the paper, and compiled and re-checked everything on his own machine.

11. Changes from v1, after the cold read

An independent reader, with no access to the author's archive, audited v1 against the Lean sources, the logs, the paper and the cited sources (30 September 2026). Verdict: **the certificate holds; no fatal error and no gap**. The reader re-derived the matrix G from the lines of the Fermat surface (zero disagreements at $m = 5, 7$), checked both $m = 5$ certificates, recomputed the Smith profiles at $m = 5, 7, 11$, and recounted every fingerprint and number. The corrections made in v2:

- **Four wrong statements, now corrected:**

1. The paper's own proof has four steps confirmed only numerically, not one (§0).
2. The paper was named as «v2, the highest version» (header, §1.3, references). Version 4 is the published one, and the Watermark certificate is cited as v2.
3. Lean identifies the relations by mixed second differences, not by polynomials of degree $\leq m - 1$ (§3.1, step 3).
4. G does not act diagonally on the overlap pullbacks. It mixes each pair, and Lean treats the pair jointly (§3.1, step 2).

- **Unclear wording, now made precise:**

- how many intermediate theorems the final check prints, and the axioms of which of them (§1);
- that the two $m = 5$ cone counts are differences of counts (§3.2);
- that the 9 GB figure is an estimate (§5);
- what the clean-rebuild log contains (§7);
- that the logs do not record the fingerprints (§8);
- how the Lean proof differs in detail from the pencil proof (§2).

- **Three further points concern the pencil proof**, now in its version 2 [DL-Pencil]:

- its status line;
- the unit binomial coefficients behind the independence in Lemma 3.1;
- the passage from the Smith form of G to V^*/V .

- **One concerns the paper:** its Abstract read as a complete proof. Version 4 says which proof is complete.

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- [Sh15] T. Shioda, *Mordell–Weil lattice of higher genus fibration on a Fermat surface*, J. Math. Sci. Univ. Tokyo **22** (2015), 443–468.
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- R. Amichis Luengo, *The Double Ladder Theorem*, 5 June 2026, version 4 of 30 September 2026; repositories *chaise-longue-theorem* (*hodge-fermat-campaign/*) and *watermark-theorem* (*papers/*).
- The Lean 4 theorem prover, <https://lean-lang.org>; the Mathlib library, <https://github.com/leanprover-community/mathlib4>.
- Aristotle, Harmonic, <https://aristotle.harmonic.fun>.